GHAZI UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE

LABORATORY ASSIGNMENT • LAB #03

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PROGRAM	**BS Computer Science**
SECTION	**D**
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Lab Task 1 · Input Statement

Question

Assume that the following variables are defined: int age; double pay; char section. Write a single cin statement that will read input into each of these variables.

Handwritten Code / Solution

Solution

TASK #1

```
#include <iostream>
using namespace std;

int main() {
    int age;
    float pay;
    char section;

    cout << "Enter age, pay and section";
    cin >> age >> pay >> section;

    cout << "Age=" << age << endl;
    cout << "Pay=" << pay << endl;
    cout << "Section=" << section << endl;

    return 0;
}
```

OUTPUT
Enter age, pay and section
Age=18
Pay=400
Section=A

```cpp
#include <iostream>
using namespace std;

int main() {
    int age;
    float pay;
    char section;

    cout << "Enter age, pay and section: ";
    cin >> age >> pay >> section;

    cout << "Age = " << age << endl;
    cout << "Pay = " << pay << endl;
    cout << "Section = " << section << endl;

    return 0;
}
```


Enter age, pay and section: 18 4000 A
Age = 18
Pay = 4000
Section = A
[Program finished]

Lab Task 2 · Expressions

Question

Complete the table by writing the value of each expression according to C++ language rules.

Handwritten Code / Solution

Solution

TASK #2

```
#include <iostream>
using namespace std;

int main() {
    cout << "20/4-2 = " << 20/4-2 << endl;
    cout << "(6+12)-9 = " << (6+12)-9 << endl;
    cout << "4+8*2 = " << 4+8*2 << endl;
    cout << "6+17%3-2 = " << 6+17%3-2 << endl;
    cout << "2+22*(9-7) = " << 2+22*(9-7) << endl;
    cout << "(8+7)*2 = " << (8+7)*2 << endl;
    cout << "(16+7)%3 = " << (16+7)%3 << endl;
    cout << "12/(10-6) = " << 12/(10-6) << endl;
    cout << "(193)*(2+2)/4 = " << (193)*(2+2)/4 << endl;

    return 0;
}
```

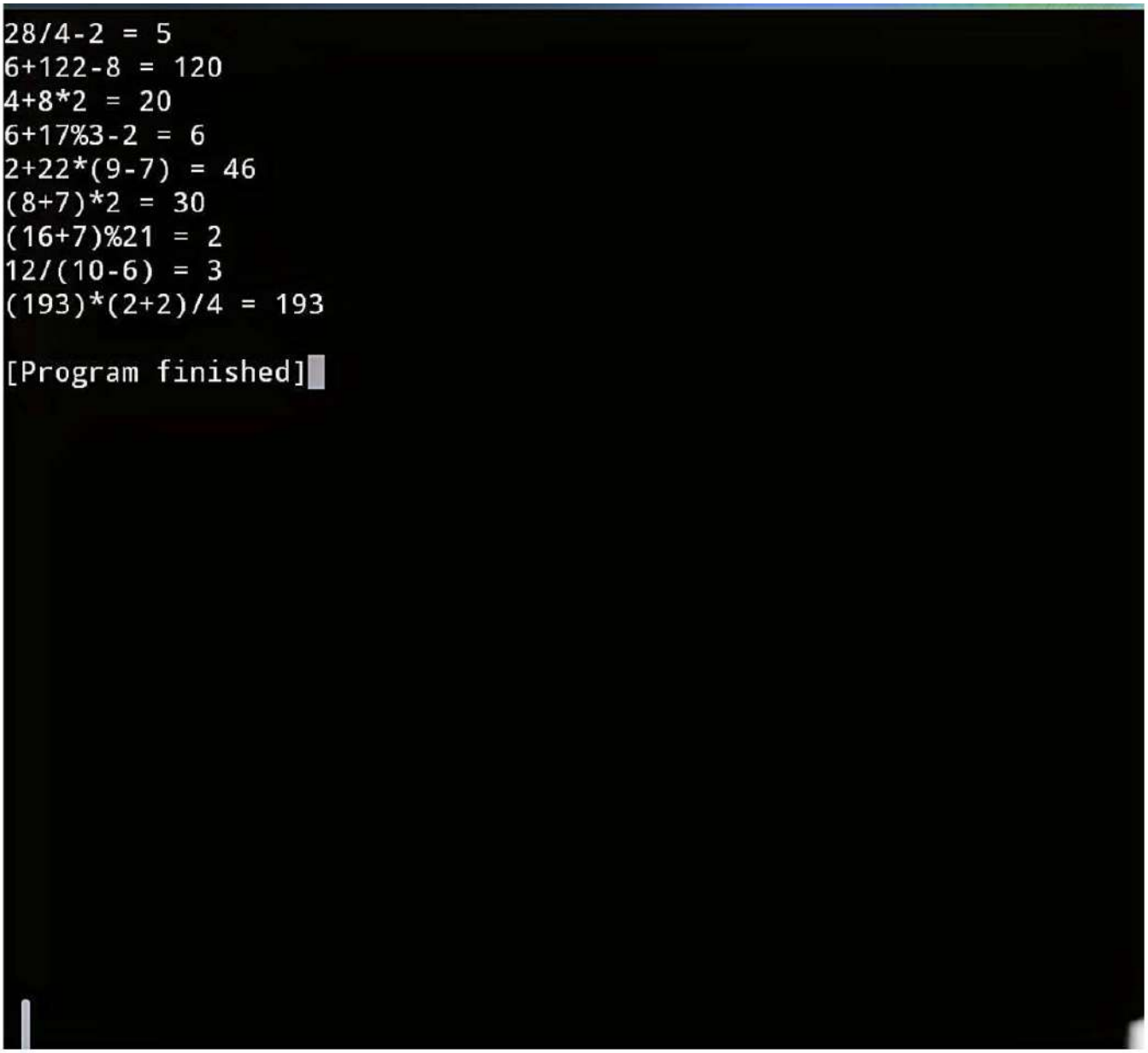
OUTPUT

```
20/4-2 = 3
(6+12)-9 = 9
4+8*2 = 20
6+17%3-2 = 6
2+22*(9-7) = 46
(8+7)*2 = 30
(16+7)%3 = 2
12/(10-6) = 3
(193)*(2+2)/4 = 193
```

```cpp
#include <iostream>
using namespace std;

int main()
{
    cout << "28/4-2 = " << 28/4-2 << endl;
    cout << "6+122-8 = " << 6+122-8 << endl;
    cout << "4+8*2 = " << 4+8*2 << endl;
    cout << "6+17%3-2 = " << 6+17%3-2 << endl;
    cout << "2+22*(9-7) = " << 2+22*(9-7) << endl;
    cout << "(8+7)*2 = " << (8+7)*2 << endl;
    cout << "(16+7)%21 = " << (16+7)%21 << endl;
    cout << "12/(10-6) = " << 12/(10-6) << endl;
    cout << "(193)*(2+2)/4 = " << (193)*(2+2)/4
<< endl;

    return 0;
}
```


28/4-2 = 5
6+122-8 = 120
4+8*2 = 20
6+17%3-2 = 6
2+22*(9-7) = 46
(8+7)*2 = 30
(16+7)%21 = 2
12/(10-6) = 3
(193)*(2+2)/4 = 193
[Program finished]

Lab Task 3 · Automatic Data Type Conversion

Question

Assume: int units; float mass; double weight; weight = mass * units. Which automatic data type conversion will take place?

Handwritten Code / Solution

Solution

#3

```
#include <iostream>
using namespace std;

int main() {
    int units = 5;
    float mass = 2.5;
    double weight;

    weight = mass * units;

    cout << "Units = " << units << endl;
    cout << "Mass = " << mass << endl;
    cout << "Weight = " << weight << endl;

    return 0;
}
```

OUTPUT
Units = 5
Mass = 2.5
Weight = 12.5

```cpp
#include <iostream>
using namespace std;

int main()
{
    int units = 5;
    float mass = 2.5;
    double weight;

    weight = mass * units;

    cout << "Units = " << units << endl;
    cout << "Mass = " << mass << endl;
    cout << "Weight = " << weight << endl;

    return 0;
}
```


Units = 5
Mass = 2.5
Weight = 12.5
[Program finished]

Lab Task 4 · Value Stored in a Variable

Question

Assume: int a, b = 2; float c = 4.2; a = b * c. What value will be stored in a?

Handwritten Code / Solution

Solution

TASK #4

```
#include <iostream>
using namespace std;

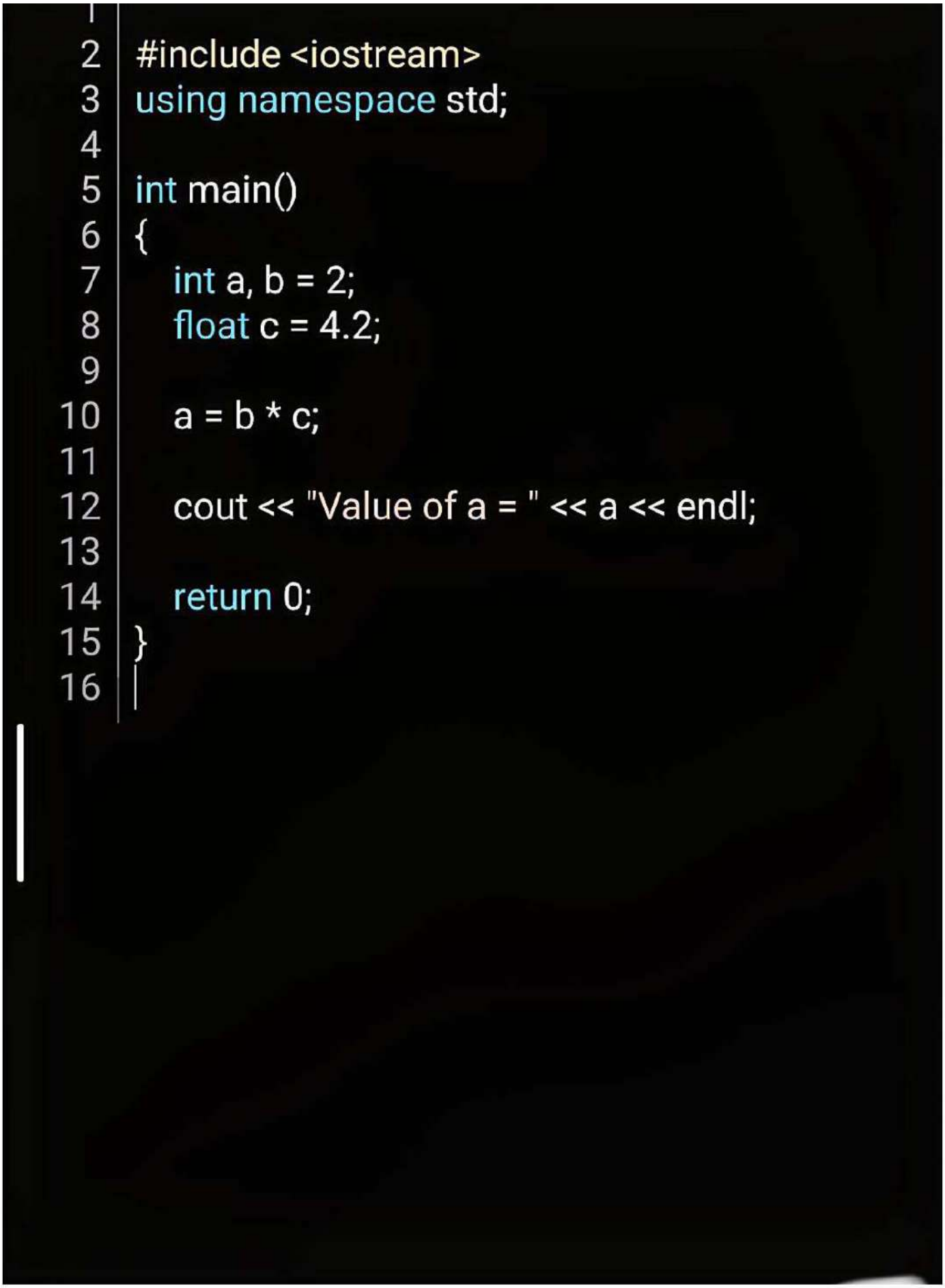
int main() {
    int a, b = 2;
    float c = 4.2;

    a = b * c;

    cout << "Value of a = " << a << endl;

    return 0;
}
```

OUTPUT
Value of a = 8



```cpp
#include <iostream>
using namespace std;

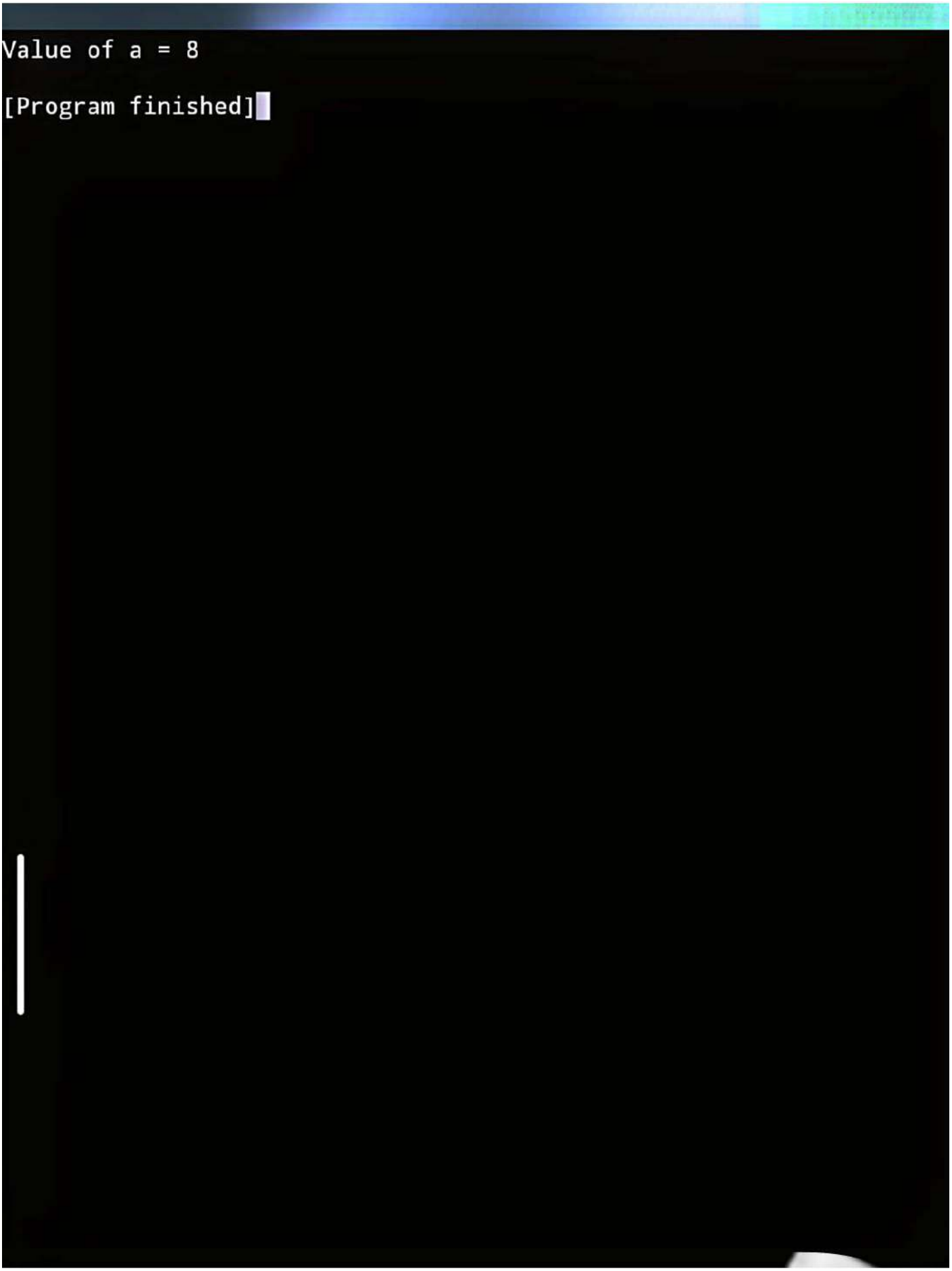
int main()
{
    int a, b = 2;
    float c = 4.2;

    a = b * c;

    cout << "Value of a = " << a << endl;

    return 0;
}
```

CamScanner


Value of a = 8
[Program finished]

Lab Task 5 · Type Casting

Question

Assume qty and salesReps are integers. Use a type cast expression to rewrite unitsEach = qty / salesReps; so that it no longer performs integer division.

Handwritten Code / Solution

Solution

#5

```
#include <iostream>
using namespace std;

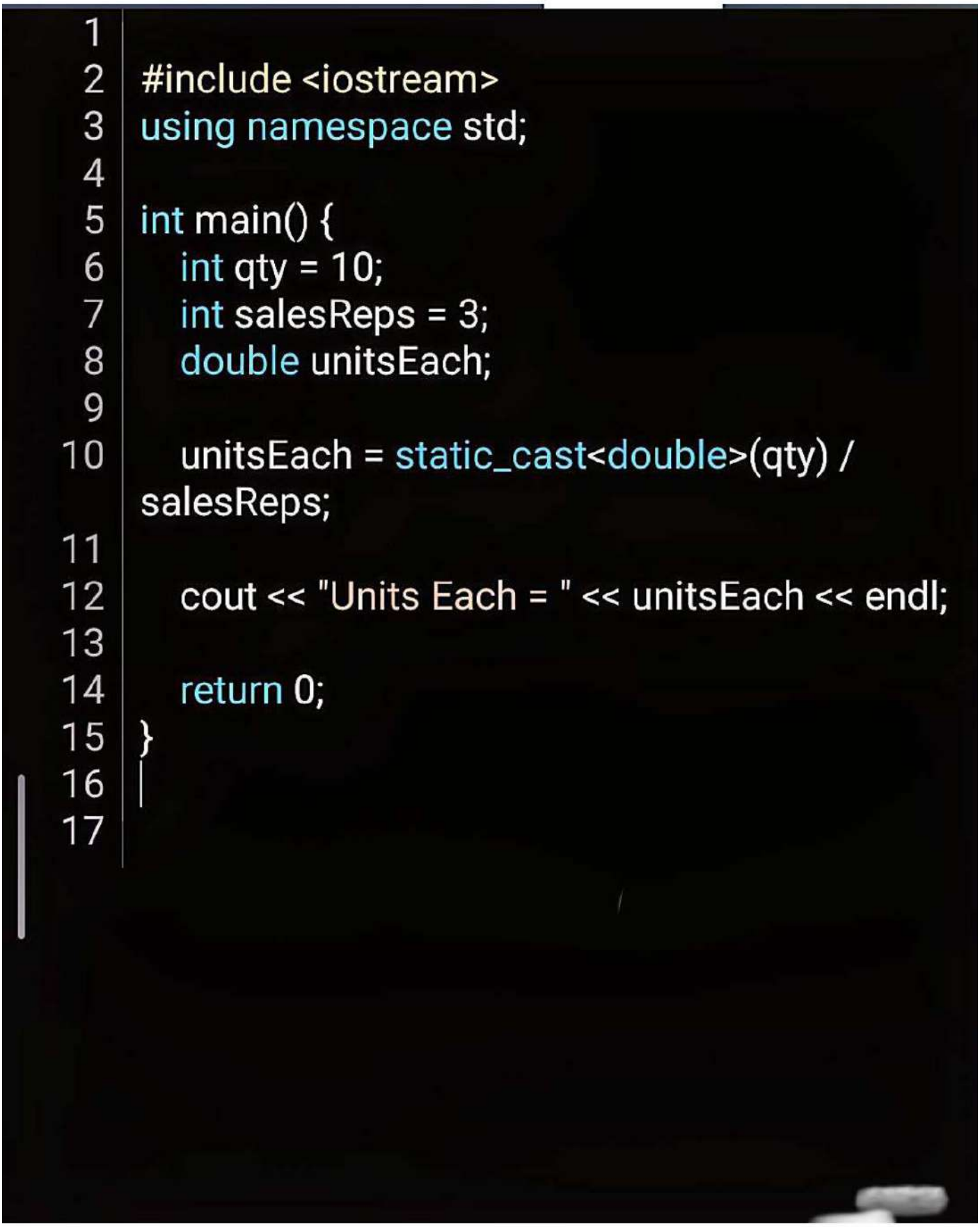
int main() {
    int qty = 10;
    int salesReps = 3;
    double unitsEach;

    unitsEach = static_cast<double>(qty) / salesReps;

    cout << "Units Each = " << unitsEach << endl;

    return 0;
}
```

OUTPUT
Units Each = 3.33333



```cpp

#include <iostream>
using namespace std;

int main() {
    int qty = 10;
    int salesReps = 3;
    double unitsEach;

    unitsEach = static_cast<double>(qty) /
salesReps;

    cout << "Units Each = " << unitsEach << endl;

    return 0;
}

```



CamScanner

Lab Task 6 · Math Tutor

Question

Write a program to display two random three-digit numbers for addition. Pause until the student is ready, then display the correct solution.

Handwritten Code / Solution

Solution

TASK #6

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

int main() {
    srand(time(0));

    int num1 = rand() % 100 + 100;
    int num2 = rand() % 100 + 100;

    cout << "Math Tutor" << endl;
    cout << "Solve this addition problem" << endl;
    cout << num1 << endl;
    cout << "+" << num2 << endl;

    cout << "Press Enter to see the answer..." << endl;
    cout << "\nCorrect Answer" << endl;
    cout << num1 << endl;
    cout << "+" << num2 << endl;
    cout << "----" << endl;
    cout << num1 + num2 << endl;

    return 0;
}
```

Solution

MATH TUTOR ? ADDITION PROBLEM

Solve the addition problem.

Press Enter to see the answer.

Correct Answer:
num1 + num2

```cpp

#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

int main()
{
    srand(time(0));

    int num1 = rand() % 900 + 100;
    int num2 = rand() % 900 + 100;

    cout << "Math Tutor" << endl;
    cout << "Solve this addition problem:" << endl;


    cout << num1 << endl;
    cout << "+ " << num2 << endl;

    cout << "\nPress Enter to see the answer...";
    cin.get();

    cout << "\nCorrect Answer:" << endl;
    cout << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "------" << endl;
    cout << num1 + num2 << endl;

    return 0;
}
```


Math Tutor
Solve this addition problem:
896
+ 111
Press Enter to see the answer...1005
Correct Answer:
896
+ 111

1007
[Program finished]

Home Task 1 · Program 1

Question

Locate and correct the errors in the provided program for adding two numbers.

Handwritten Code / Solution

Solution

HOME TASK 1 ? PROGRAM 1

```
#include <iostream>
using namespace std;

int main() {
    int number1, number2, sum;

    cout << "Enter first number: ";
    cin >> number1;

    cout << "Enter second number: ";
    cin >> number2;

    sum = number1 + number2;

    cout << "The sum of the two numbers is: "
         << sum << endl;

    return 0;
}
```

Solution

```
Enter a number: 3
Enter another number: 6
The sum of the two numbers is: 9
```

PROGRAM 1

```
#include <iostream>
using namespace std;

int main() {
    int number1, number2;
    int sum;

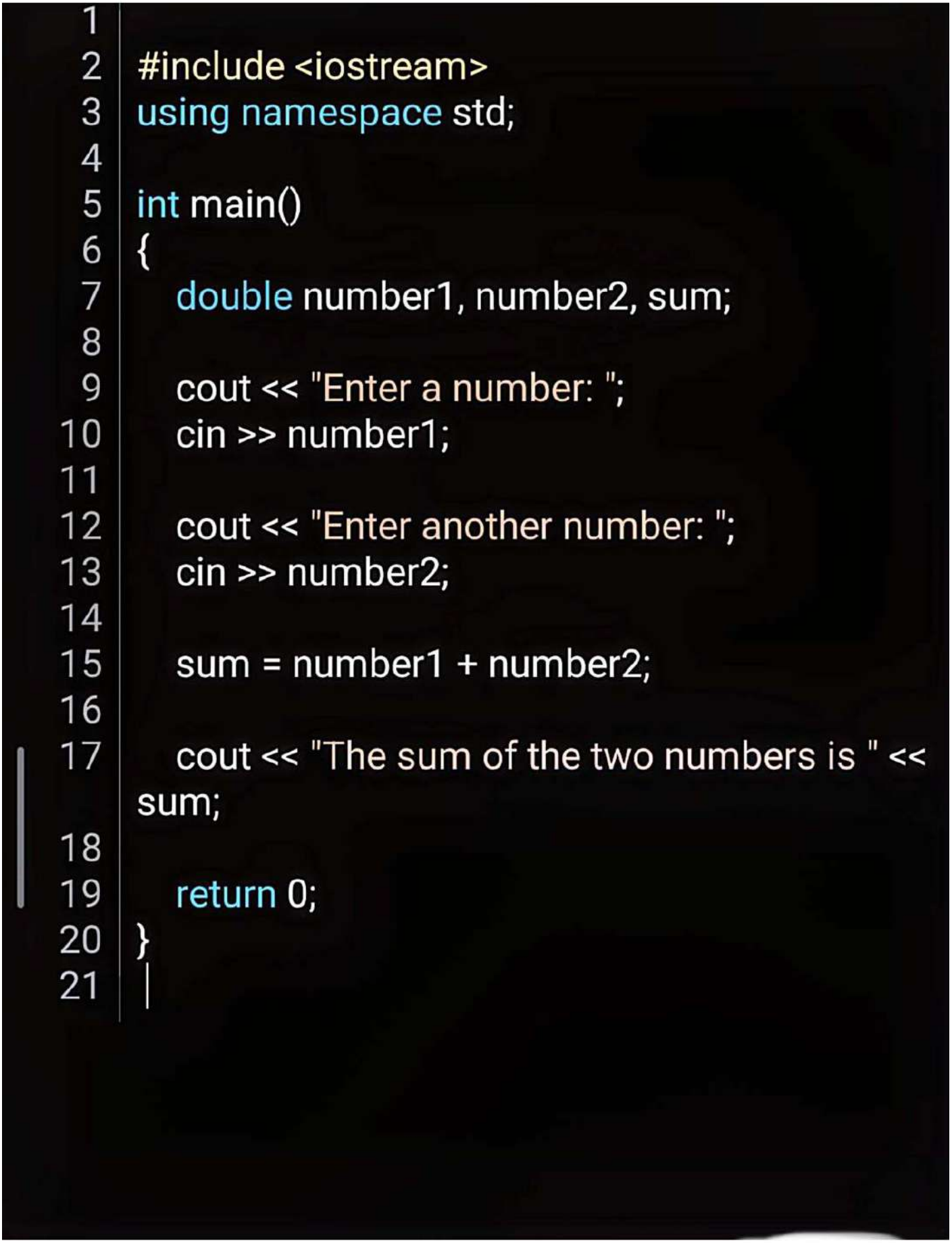
    cout << "Enter first number: ";
    cin >> number1;

    cout << "Enter second number: ";
    cin >> number2;

    sum = number1 + number2;

    cout << "The sum of the two numbers is: "
         << sum << endl;

    return 0;
}
```



```cpp

#include <iostream>
using namespace std;

int main()
{
    double number1, number2, sum;

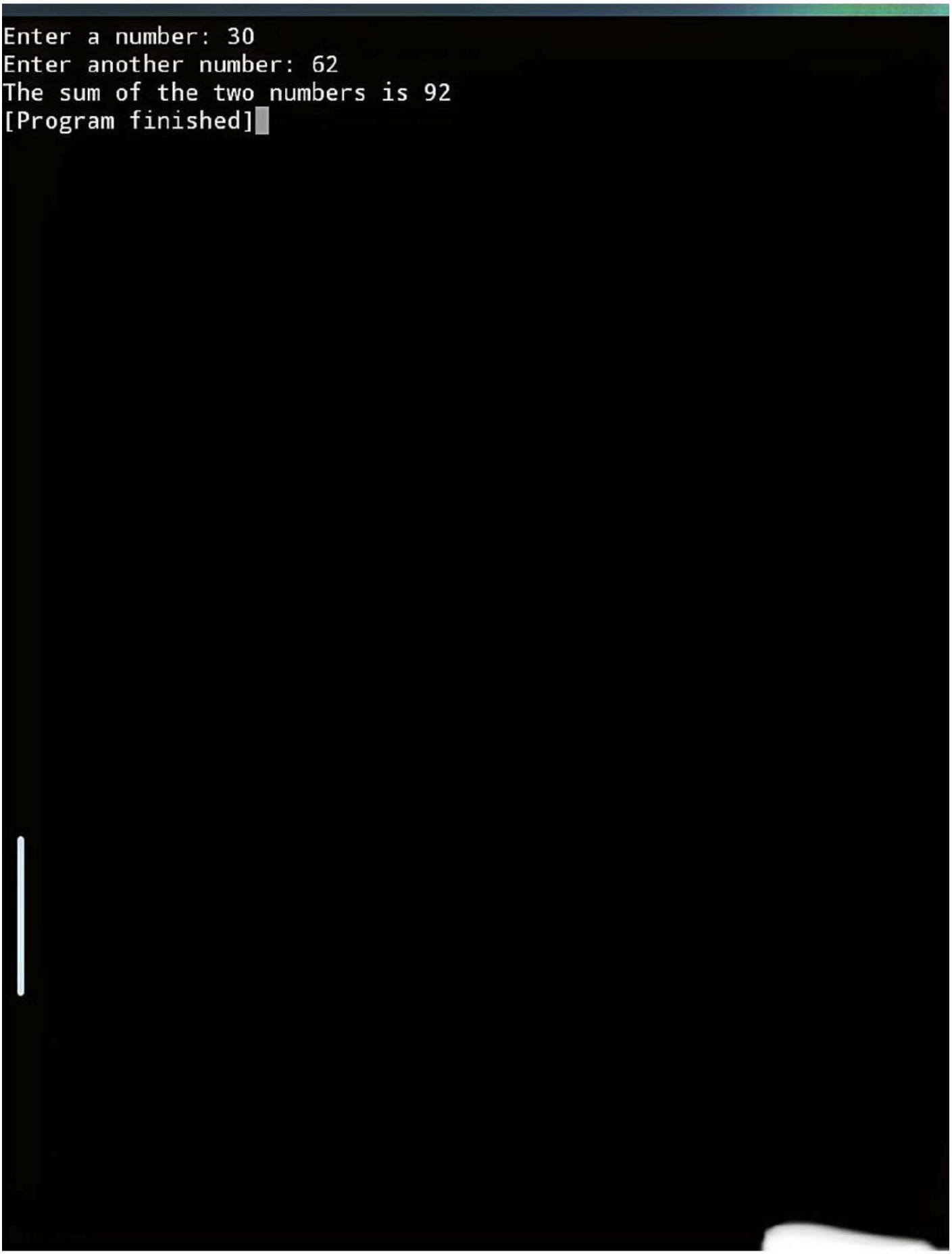
    cout << "Enter a number: ";
    cin >> number1;

    cout << "Enter another number: ";
    cin >> number2;

    sum = number1 + number2;

    cout << "The sum of the two numbers is " <<
sum;

    return 0;
}
```


Enter a number: 30
Enter another number: 62
The sum of the two numbers is 92
[Program finished]

Home Task 1 · Program 2

Question

Locate and correct the errors in the provided program for dividing two numbers.

Handwritten Code / Solution

Solution

```
Enter a number: 22
Enter another number: 6
The answer is: 3.2

PROGRAM 2

#include <iostream>
using namespace std;

int main() {
    int number1, number2;
    float quotient;

    cout << "Enter a number: ";
    cin >> number1;

    cout << "Enter another number: ";
    cin >> number2;

    quotient = static_cast<float>(number1) / number2;

    cout << "The quotient of the two numbers is: "
         << quotient << endl;

    return 0;
}
```

Solution

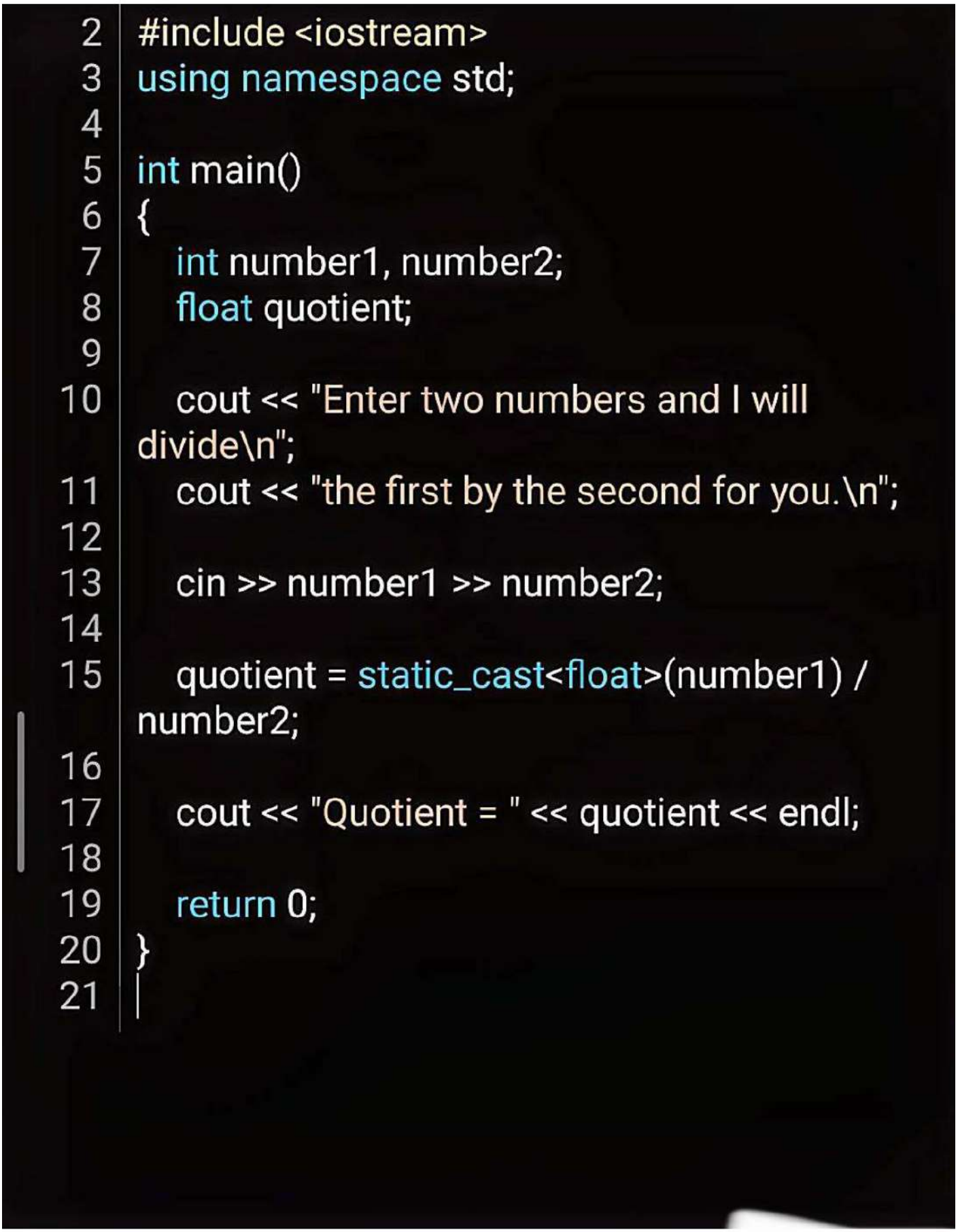
```
cout << "The quotient is: " << quotient << endl;
return 0;
}
```

OUTPUT

```
Enter the first number: 22
Enter the second number: 6

Quotient = 22 / 6

= 3.66667
```



```cpp
#include <iostream>
using namespace std;

int main()
{
    int number1, number2;
    float quotient;

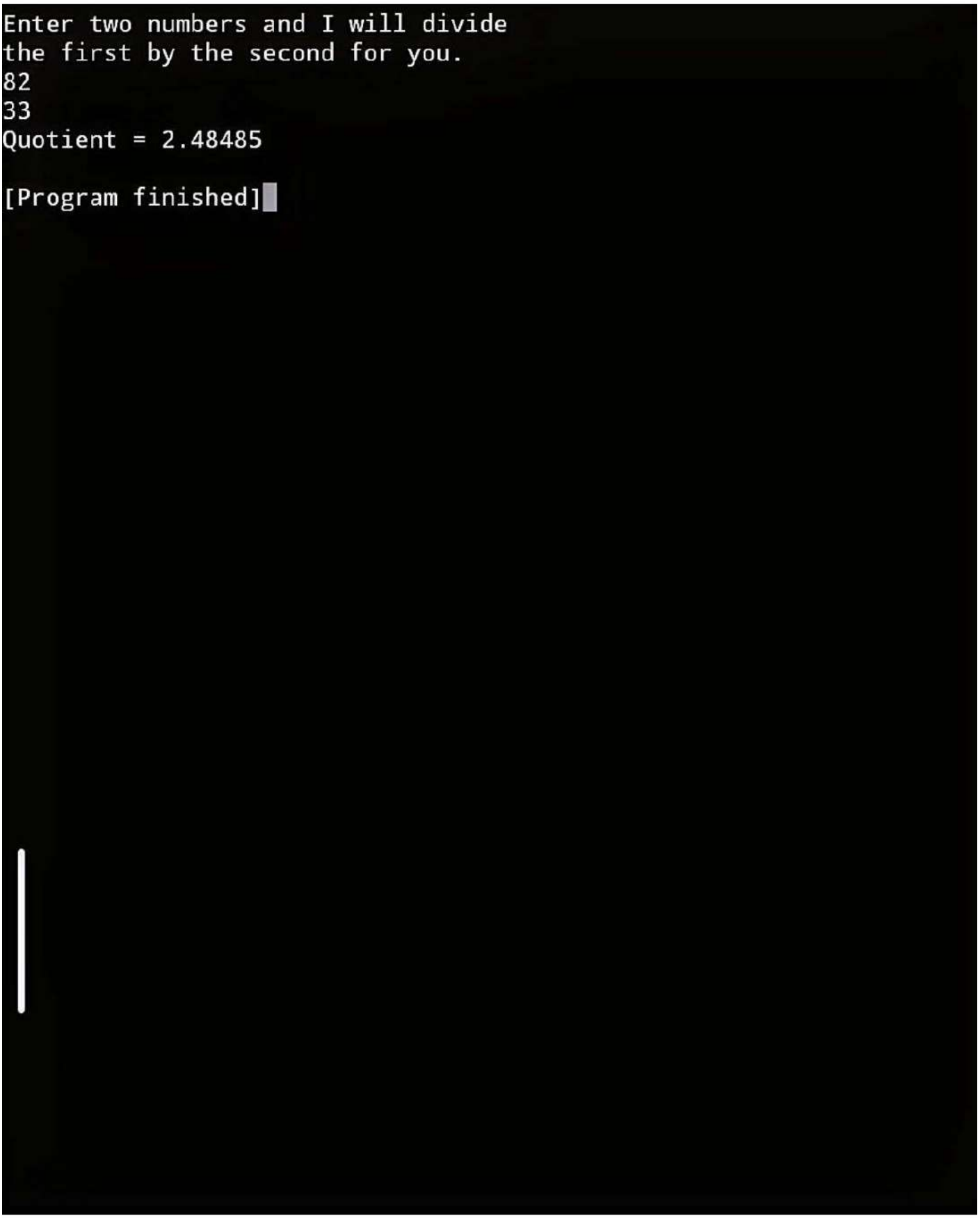
    cout << "Enter two numbers and I will divide\n";
    cout << "the first by the second for you.\n";

    cin >> number1 >> number2;

    quotient = static_cast<float>(number1) / number2;

    cout << "Quotient = " << quotient << endl;

    return 0;
}
```


Enter two numbers and I will divide
the first by the second for you.
82
33
Quotient = 2.48485
[Program finished]

Home Task 2 · Average of Values

Question

Store 28, 32, 37, 24, and 33 in five variables. Calculate the sum, divide by five, and display the average.

Handwritten Code / Solution

Solution

```
#include <iostream>
using namespace std;

int main() {
    int number1 = 28;
    int number2 = 32;
    int number3 = 37;
    int number4 = 24;
    int number5 = 33;
    int sum;

    cout << "Average: " << endl;

    sum = number1 + number2 + number3 + number4 + number5;

    cout << "Sum = " << sum << endl;
    cout << "Average = " << sum / 5.0 << endl;

    return 0;
}
```

Solution

OUTPUT

Sum = 154
Average = 30.8

```cpp

#include <iostream>
using namespace std;

int main()
{
    int num1 = 28;
    int num2 = 32;
    int num3 = 37;
    int num4 = 24;
    int num5 = 33;

    int sum;
    float average;

    sum = num1 + num2 + num3 + num4 +
num5;

    average = sum / 5.0;

    cout << "Sum = " << sum << endl;
    cout << "Average = " << average << endl;

    return 0;
}

```


Sum = 154
Average = 30.8
[Program finished]